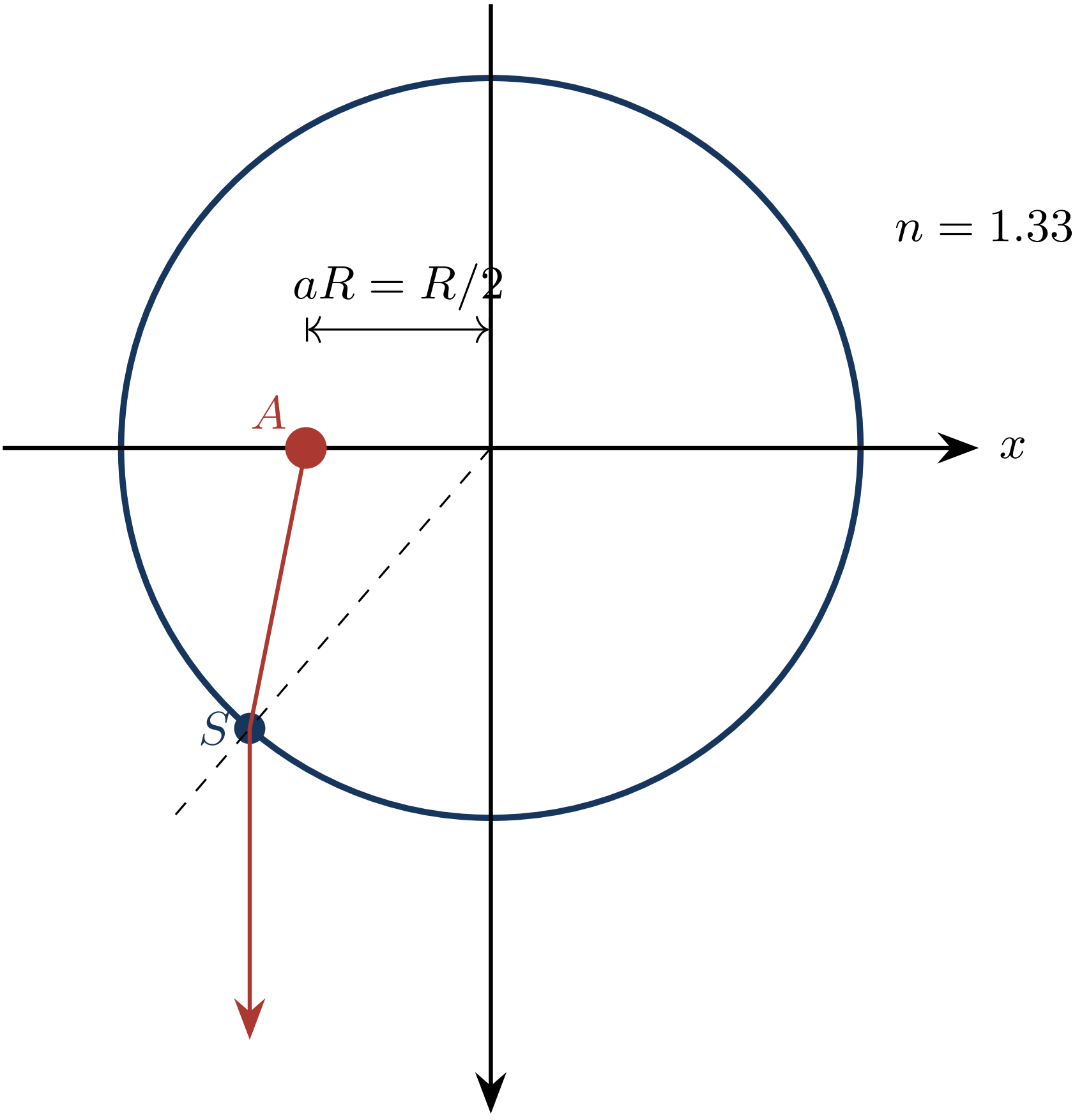
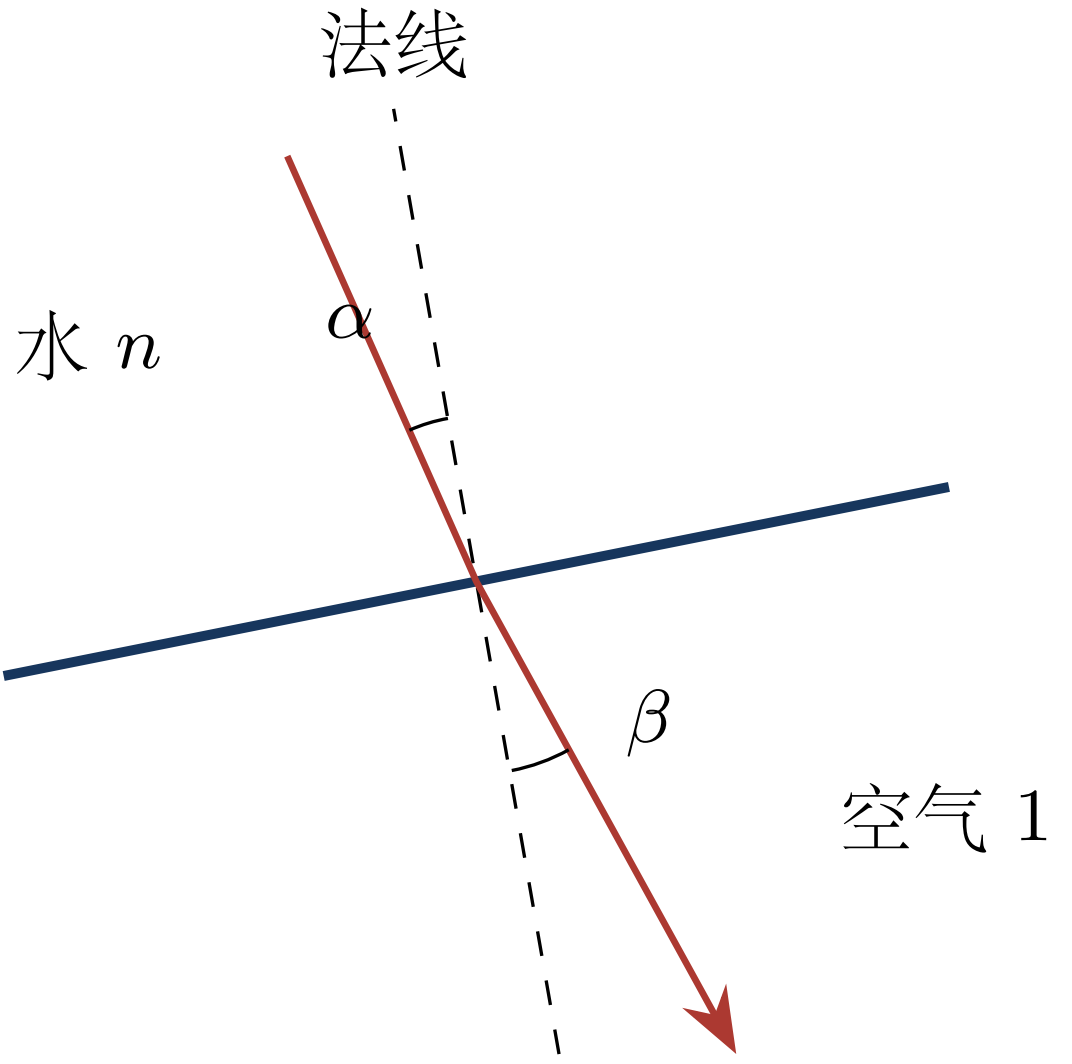
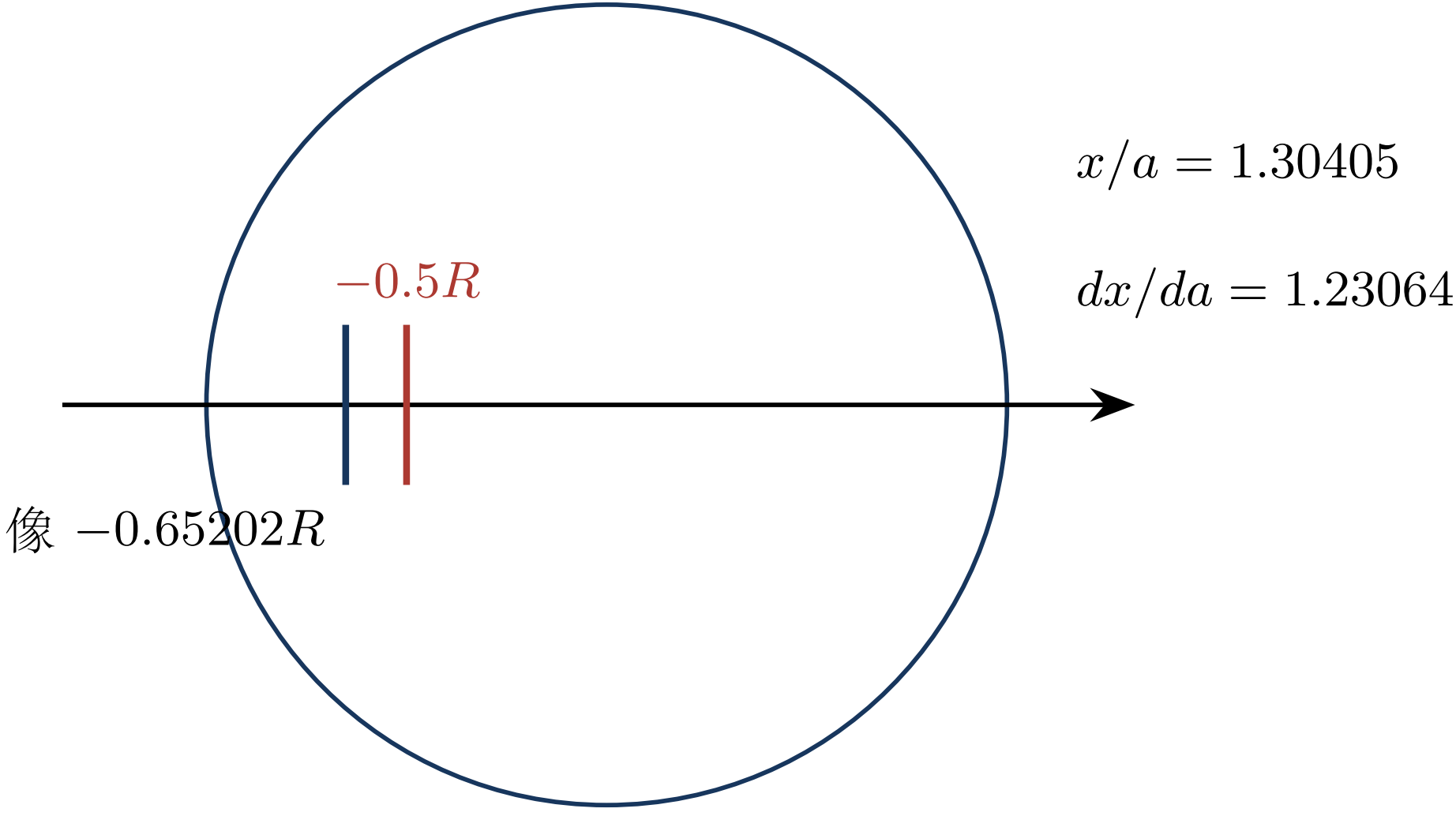
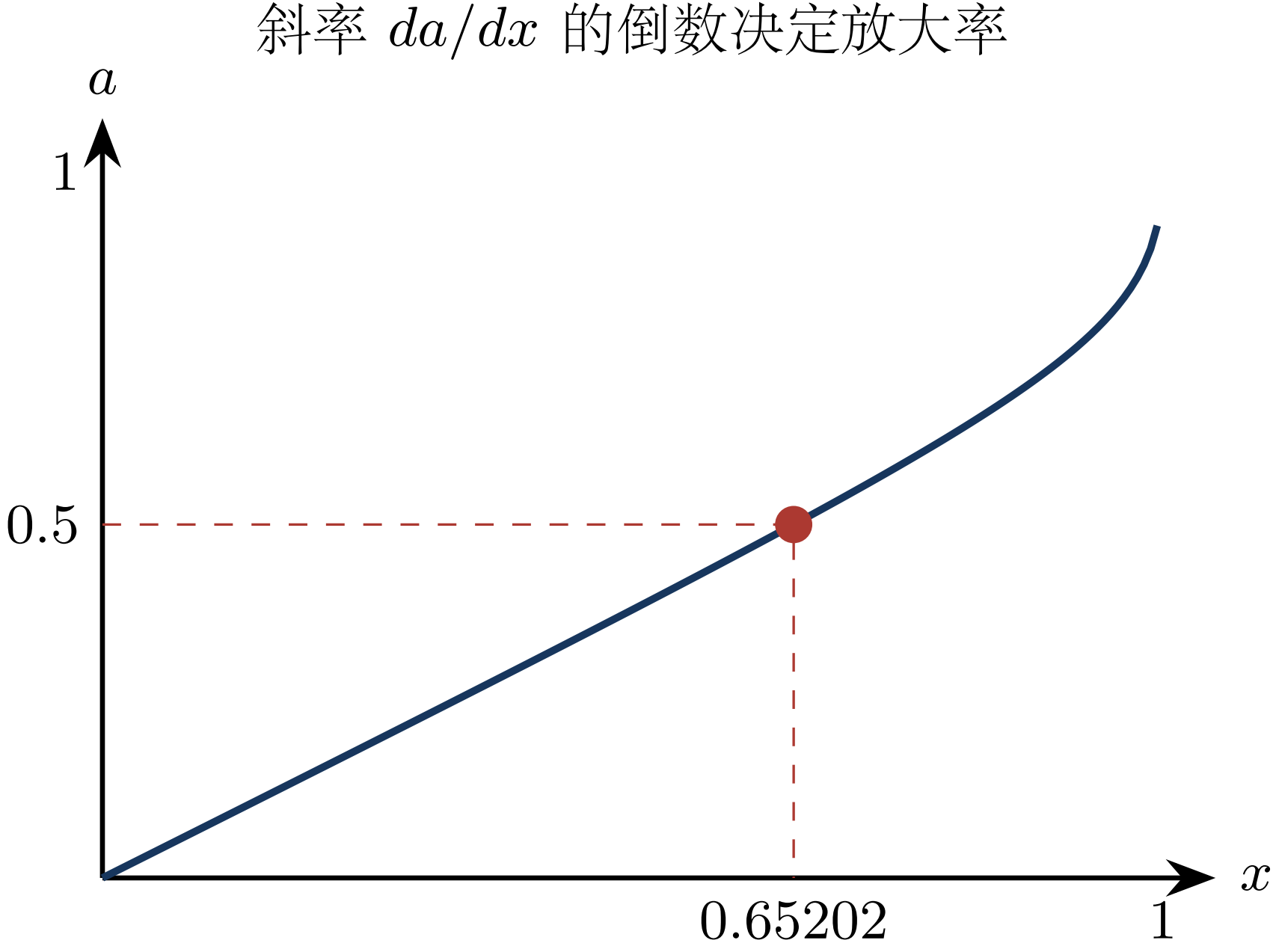

$n = 1.33$
$aR = R/2$
A
x
S
观察者 →



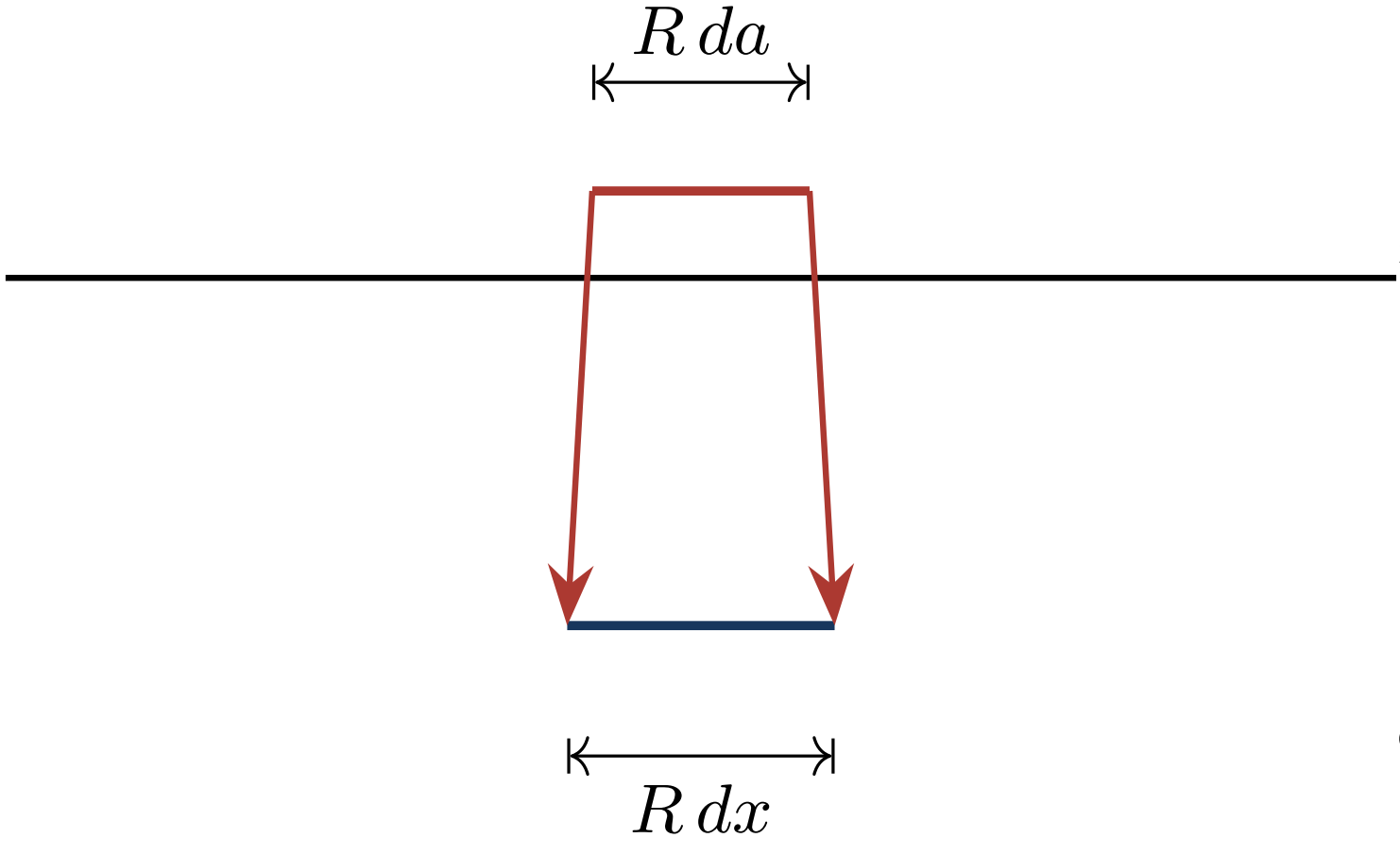
$$n \sin \alpha = \sin \beta = x$$



像位置的比例与宽度放大率不同


斜率 da/dx 的倒数决定放大率
a
1
0.5
x
0.65202
1

$$M = dx/da \simeq 1.231$$



真实细宽度

$dw = R\,da$

图像细宽度

$dw' = R\,dx$